

SERVICENOW CAPSTONE PROJECT REPORT

**AN EMPLOYEE REQUESTS INSTALLATION OF LICENSED
SOFTWARE THROUGH THE SERVICE CATALOG**

A PROJECT REPORT

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MAY 2026

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

Organizations rely on various software applications to perform daily business operations efficiently. In many organizations, employees request software installations through emails, phone calls, or manual forms. These traditional approaches often result in delayed approvals, poor request tracking, inconsistent communication, and increased workload for IT support teams.

The Software Installation Request Automation using ServiceNow project addresses these challenges by automating the complete software request lifecycle through the ServiceNow platform. The solution utilizes Service Catalog, Workflow Editor, UI Policies, Catalog Tasks, Notifications, and ServiceNow database tables (REQ, RITM, and SCTASK) to streamline software request management.

Employees can submit software installation requests through a self-service portal by providing details such as Software Name, Version, Business Justification, License Type, and Urgency. The system validates the submitted information, automatically initiates the approval workflow, generates catalog tasks for the Software Support Team, sends email notifications, and allows users to track their requests until completion.

The automated solution reduces manual intervention, improves transparency, enhances operational efficiency, and provides a standardized process for handling software installation requests across the organization.

1.2 Purpose

The primary purpose of this project is to automate the software installation request process using the ServiceNow platform and eliminate manual request handling.

The project aims to:

- Develop a centralized Service Catalog for software installation requests.
- Automate request validation using UI Policies and Business Rules.
- Configure approval workflows for software requests.
- Automatically generate Catalog Tasks for the IT Software Support Team.

- Enable real-time tracking of requests through REQ, RITM, and SCTASK records.
- Send automated email notifications during every stage of the request lifecycle.
- Improve operational efficiency by reducing manual processing and approval delays.
- Enhance the user experience through a simple self-service portal.
- Provide a scalable solution that can be extended to automate additional IT services such as hardware requests, access requests, and employee onboarding.

CHAPTER 2

IDEATION PHASE

2.1 Problem Statement

In many organizations, employees request software installations through emails, paper forms, or direct communication with the IT support team. These traditional methods often lead to delayed approvals, inconsistent communication, manual task assignment, and limited visibility into the request status. As the number of requests increases, managing them manually becomes time-consuming and error-prone.

To overcome these challenges, this project proposes an automated Software Installation Request System using the ServiceNow platform. The solution enables employees to submit software requests through the Service Catalog while automating approvals, catalog task creation, notifications, and request tracking. This approach reduces manual effort, improves transparency, and ensures a standardized software request process.



Fig 2.1 Problem Statement

2.2 Empathy Map Canvas

The empathy map was created to understand the needs, expectations, frustrations, and behaviors of the primary user, i.e., the employee requesting software installation.

Customer : Employee requesting software installation

Think & Feel

- I need software quickly to complete my work.
- The request process should be simple.

- I want to know the status of my request.
- I expect timely approval and installation.

See

- Different software requirements across departments.
- Delays in manual approval processes.
- Multiple communication channels for software requests.
- Lack of visibility into request progress.

Say & Do

- Submit software installation requests.
- Follow up with IT regarding request status.
- Provide business justification for software.
- Check notifications and request updates.

Hear

- Recommendations from managers.
- Guidance from the IT support team.
- Organizational software policies.
- Feedback from colleagues regarding software availability.

Pain Points

- Manual approval delays.
- No centralized request tracking.
- Repetitive communication with IT.
- Slow software installation process.

Gains

- Automated request submission.
- Faster approvals.
- Real-time request tracking.
- Quick software installation.
- Better communication through notifications.



Fig 2.2 Empathy Map

2.3 Brainstorming

Several solution ideas were analyzed before selecting the final approach for the project.

Ideas Considered

- Manual email-based software request system.
- Web-based request management application.
- Ticket management system.
- ServiceNow Service Catalog automation.
- Workflow-based approval system.
- Automated catalog task generation.

Selected Idea

The final solution selected was Software Installation Request Automation using ServiceNow because it provides:

- Centralized software request management.
- Automated approval workflows.
- Dynamic form validation using UI Policies.
- Automatic catalog task creation and Email notifications.
- Real-time request tracking using REQ, RITM, and SCTASK tables.
- Easy scalability for future IT service automation.

Idea Prioritization

The selected solution was prioritized based on the following factors:

- High business value.
- Easy integration with ServiceNow.
- Reduced manual effort.
- Improved employee experience.
- Faster service delivery.
- Better process transparency.
- Future scalability.

Outcome of Ideation Phase

The ideation phase helped identify the key challenges faced during software request processing and evaluate multiple solution approaches. Based on feasibility, automation capability, scalability, and user experience, the ServiceNow-based solution was selected. This phase established the foundation for the subsequent requirement analysis, system design, implementation, testing, and deployment phases of the project.



Fig 2.3 Brainstorm & Idea prioritization

CHAPTER 3

REQUIREMENT ANALYSIS

The Requirement Analysis phase focuses on understanding the business needs, identifying system users, defining functional and non-functional requirements, analyzing the request flow, and selecting the appropriate technologies for implementing the solution. During this phase, the existing software request process was studied, and a ServiceNow-based automated solution was proposed to improve efficiency, transparency, and request management.

3.1 Customer Journey Map

The Customer Journey Map illustrates the complete experience of an employee from identifying the need for software until the request is fulfilled. It helps understand user interactions, expectations, and opportunities for process improvement.

Stage 1 – Need Identification

The employee identifies the need for software to perform assigned tasks or project work.

Customer Goal

- Obtain the required software quickly.

Pain Point

- Existing request process is manual and time-consuming.

Stage 2 – Request Submission

The employee logs into the ServiceNow Service Portal and submits a Software Installation Request through the Service Catalog.

Activities

- Select Software Installation Request
- Enter Software Name
- Enter Version
- Provide Business Justification
- Select License Type
- Submit Request

3.2 Solution Requirements

The project requirements were classified into Functional Requirements and Non-Functional Requirements.

Functional Requirements

- Software Installation Request submission through Service Catalog.
- Mandatory field validation.
- Dynamic UI behavior using UI Policies.
- Automated approval workflow.
- Catalog Task generation.
- Automatic task assignment.
- Email notification automation.
- Request tracking using REQ, RITM, and SCTASK.
- Update Set migration.
- Administrative management.

Non-Functional Requirements

- User-friendly Service Portal interface.
- Secure access using role-based permissions.
- Reliable workflow execution.
- Fast request processing.
- High availability using ServiceNow Cloud.
- Scalable architecture for future IT services.

3.3 Data Flow Diagram

The Data Flow Diagram (DFD) represents how data flows through the Software Installation Request Automation system.

The process begins when an employee submits a software request through the Service Catalog. The request is validated and processed by the Workflow Engine. Based on the approval conditions, the request is forwarded to the Manager or Software Asset Team. Once approved, the workflow automatically creates a Catalog Task for the Software Support Team. All request

information is stored in the REQ, RITM, and SCTASK tables. Throughout the process, automated notifications keep users informed until the request is completed.

Major Components

- Employee (Requester)
- Service Portal
- Service Catalog
- Workflow Editor
- Approval Process
- Catalog Task
- Software Support Team
- REQ Table
- RITM Table
- SCTASK Table
- Notification Engine

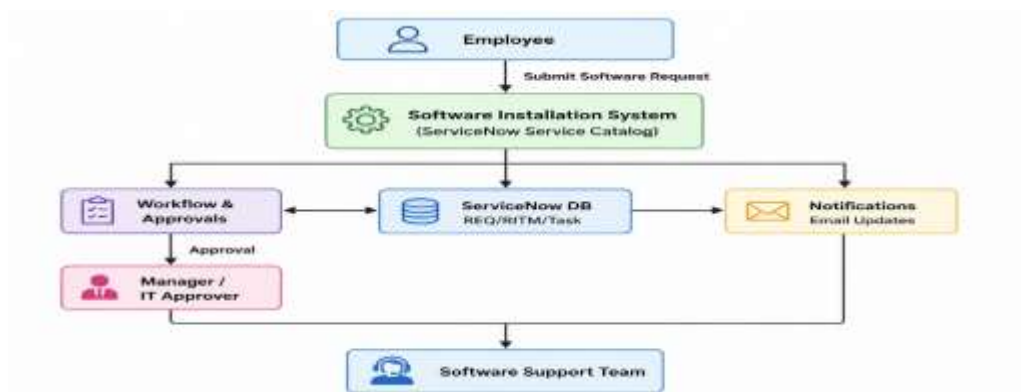


Fig 3.3 Data Flow Diagram

3.4 Technology Stack

The Software Installation Request Automation project was developed entirely on the ServiceNow platform using its built-in IT Service Management capabilities.

Component	Technology Used
Platform	ServiceNow Personal Developer Instance (PDI)
User Interface	Service Portal
Catalog Management	Service Catalog
Workflow Automation	Workflow Editor
Process Automation	Flow Designer
Form Validation	UI Policies
Server-side Logic	Business Rules
Database	ServiceNow Tables (REQ, RITM, SCTASK)
Task Management	Catalog Task (SCTASK)
Notifications	Email Notifications
Security	Role-Based Access Control (RBAC)
Migration	Update Sets
Testing	User Acceptance Testing (UAT)

Requirement Analysis Summary

The Requirement Analysis phase provided a clear understanding of the existing software request process and identified opportunities for automation. Functional requirements defined the core features of the application, while non-functional requirements ensured security, scalability, reliability, and usability. The customer journey, data flow, and technology stack established a strong foundation for the design and implementation phases. The analysis confirmed that ServiceNow provides a comprehensive platform for automating software request management through Service Catalog, Workflow Editor, Flow Designer, UI Policies, Business Rules, and integrated database tables.

CHAPTER 4

PROJECT DESIGN

The Project Design phase focused on transforming the identified business requirements into a structured technical solution. During this phase, the proposed solution was designed to automate the complete software installation request lifecycle using ServiceNow's built-in capabilities. The design includes the problem–solution fit analysis, proposed solution, and overall system architecture.

4.1 Problem–Solution Fit

The Problem–Solution Fit ensures that the proposed solution effectively addresses the problems experienced by employees and IT support teams during the software request process.

Existing Problems

Organizations often rely on manual methods such as emails, phone calls, or paper forms for software installation requests. These methods create several challenges, including:

- Delayed approval process.
- Manual task assignment.
- Lack of request status visibility.
- Poor communication between employees and IT support.
- Increased administrative workload.
- Inconsistent request documentation.

Proposed Solution

The Software Installation Request Automation solution utilizes the ServiceNow platform to automate the entire request lifecycle.

The solution provides:

- Self-service software request submission through the Service Catalog.
- Automated validation using UI Policies and Business Rules.
- Workflow-based approval automation.
- Automatic Catalog Task creation for the Software Support Team.
- Email notifications at every stage.

- Real-time request tracking using REQ, RITM, and SCTASK.
- Automatic request closure after successful fulfillment.

Benefits Achieved

- Reduced manual effort.
- Faster request processing.
- Improved transparency.
- Better user experience.
- Standardized request handling.
- Increased operational efficiency.



Fig 4.1 Problem Solution Fit

4.2 Proposed Solution

The proposed solution is an automated Software Installation Request Management System developed on the ServiceNow platform.

Employees submit software installation requests through the Service Portal by providing essential information such as:

- Software Name
- Software Version
- Business Justification
- License Type

- Urgency

After submission, the system validates the entered information and automatically initiates the approval workflow. Depending on the request type, approvals are routed to the Manager or Software Asset Team.

Once approved, the workflow automatically creates a Catalog Task (SCTASK) and assigns it to the Software Support Team for software installation. The requester receives automated notifications throughout the process and can monitor the request status using the REQ, RITM, and SCTASK records.

Key Features

- Service Catalog-based software request.
- Dynamic form behavior using UI Policies.
- Business Rule validation.
- Workflow automation.
- Approval management.
- Catalog Task automation.
- Email notifications.
- Request tracking.
- Update Set migration support.

Expected Outcomes

- Faster approval process.
- Improved IT service delivery.
- Reduced manual intervention.
- Better request visibility.
- Increased customer satisfaction.
- Scalable IT service automation platform.

4.3 Solution Architecture

The proposed solution follows a layered architecture implemented on the ServiceNow cloud platform.

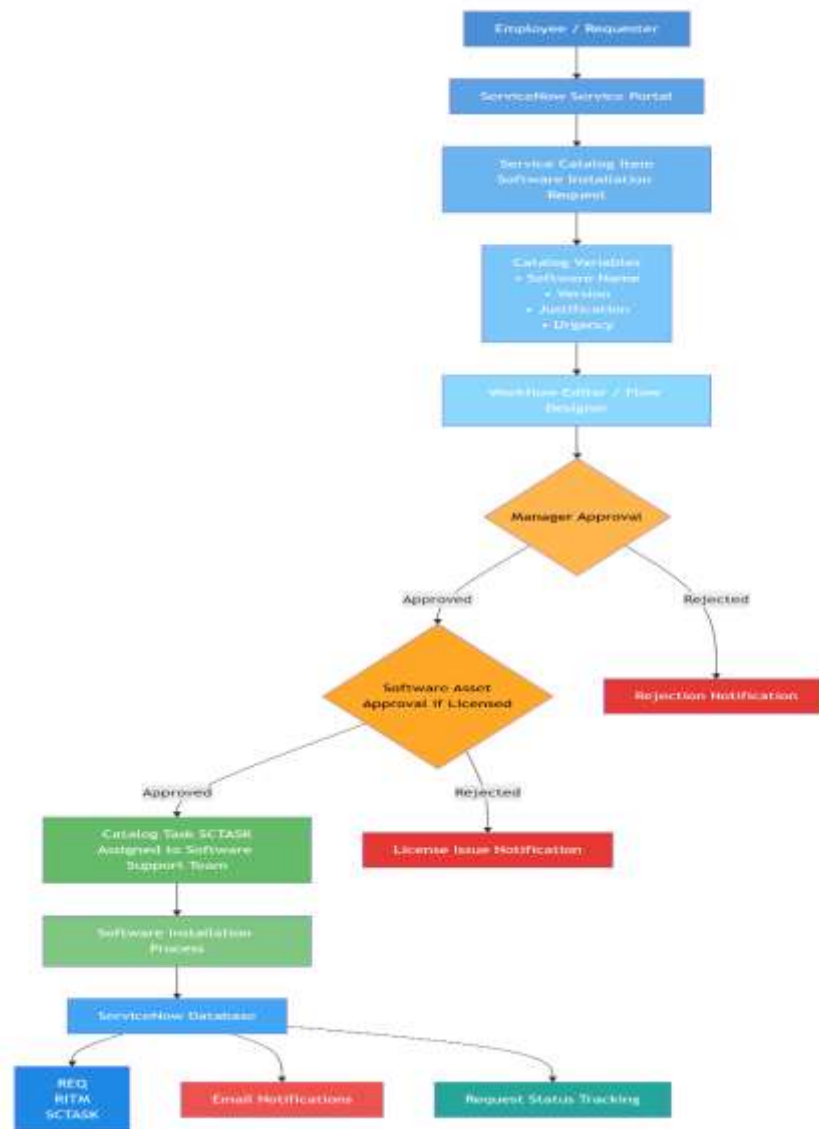


Fig 4.3 Solution Architecture

Architecture Overview

The system consists of the following major components:

User Layer

Employees access the Service Portal and submit software installation requests.

Application Layer

The Service Catalog captures user input through catalog variables such as Software Name, Version, Business Justification, License Type, and Urgency.

UI Policies and Business Rules validate the submitted information before processing.

Workflow Editor and Flow Designer automate the approval process, task generation, and request lifecycle.

Processing Layer

The workflow routes requests to:

- Manager Approval
- Software Asset Team Approval (for licensed software)
- Software Support Team

Catalog Tasks are automatically generated and assigned for software installation.

Data Layer

All request information is maintained within ServiceNow database tables.

Primary tables include:

- Request (REQ)
- Requested Item (RITM)
- Catalog Task (SCTASK)

These tables store request details, approval status, fulfillment information, and workflow history.

Notification Layer

Email Notifications automatically inform:

- Requester
- Approver
- Software Support Team

Notifications are triggered during request submission, approval, task assignment, and completion.

Solution Workflow

The overall solution workflow is as follows:

Employee → Service Portal → Service Catalog → Catalog Variables → UI Policies & Business Rules → Workflow Editor → Manager Approval → Software Asset Team Approval (if

required) → Catalog Task Creation → Software Support Team → Software Installation → REQ/RITM/SCTASK Update → Email Notification → Request Completion

Design Advantages

The proposed architecture offers several advantages:

- Modular and maintainable design.
- Cloud-based deployment using ServiceNow.
- Automated workflow execution.
- Secure role-based access control.
- Centralized request management.
- High scalability for future IT services.
- Easy maintenance through Update Sets.
- Improved employee experience through automation.

Project Design Summary

The Project Design phase successfully transformed the business requirements into a scalable and automated ServiceNow solution. The architecture integrates Service Catalog, Workflow Editor, Flow Designer, UI Policies, Business Rules, Notifications, and ServiceNow database tables to automate software installation requests. The designed solution minimizes manual effort, improves operational efficiency, enhances transparency, and provides a robust foundation for future IT service automation.

CHAPTER 5

PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Project planning is one of the most important stages of software development, as it provides a structured approach for completing the project within the planned timeline. During this phase, the overall scope of the **Software Installation Request Automation using ServiceNow** project was defined by identifying project objectives, user requirements, development activities, testing strategy, and deployment plan.

The project was planned following the **Agile Scrum methodology**, where the development process was divided into multiple sprint cycles. Each sprint consisted of specific user stories and functional requirements that were implemented, tested, and reviewed before moving to the next sprint. Story points were assigned to estimate the effort required for each task, helping in effective workload distribution and progress monitoring.

The planning process included the following activities:

- Requirement analysis and project scope identification.
- Identification of ServiceNow tables (REQ, RITM, SCTASK).
- Service Catalog planning.
- Catalog variable identification.
- Workflow and approval planning.
- UI Policy and Business Rule planning.
- Catalog Task planning.
- Notification planning.
- Testing strategy preparation.
- Update Set migration planning.
- Documentation and deployment planning.

The project schedule was prepared to ensure that every development activity was completed within the specified timeline while maintaining quality and meeting project objectives.

Project Planning Activities :

Phase	Planned Activities
Requirement Analysis	Study existing software request process, identify user requirements, and define project scope.
Design	Design Service Catalog, Workflow, Approval Process, Data Flow Diagram, and Solution Architecture.
Development	Configure Service Catalog, Catalog Variables, Workflow Editor, UI Policies, Business Rules, Notifications, and Catalog Tasks.
Testing	Perform functional testing, workflow validation, User Acceptance Testing (UAT), and defect fixing.
Deployment	Export and migrate Update Sets, validate configurations, and deploy the solution.
Documentation	Prepare technical documentation, user manual, architecture diagrams, testing reports, and final project report.

Planning Outcome

The project planning phase established a clear roadmap for implementing the Software Installation Request Automation solution. By adopting Agile Scrum practices and structured sprint planning, the project was completed systematically, ensuring timely delivery, effective resource utilization, and successful achievement of all functional and non-functional requirements.

CHAPTER 6

FUNCTIONAL & PERFORMANCE TESTING

Testing is an essential phase in the software development lifecycle that ensures the developed solution meets the specified functional and non-functional requirements. The Software Installation Request Automation using ServiceNow project underwent comprehensive testing to verify the correctness, reliability, usability, and overall performance of the implemented solution.

Testing was performed after the completion of the development phase to ensure that every component of the software request lifecycle functioned as expected. Both positive and negative test scenarios were executed to validate the application's behavior under different conditions.

Functional Testing

Functional testing was conducted to verify that every feature of the Software Installation Request Automation system performs according to the specified requirements.

The following functionalities were successfully tested:

- Software Installation Request submission through Service Catalog.
- Mandatory field validation using UI Policies and Business Rules.
- Dynamic display of form fields based on License Type.
- Automatic approval workflow execution.
- Manager Approval process.
- Software Asset Team Approval (for licensed software).
- Automatic Catalog Task (SCTASK) creation.
- Task assignment to the Software Support Team.
- Email notification generation.
- Request tracking through REQ, RITM, and SCTASK.
- Automatic request completion after task fulfillment.
- Update Set migration between ServiceNow instances.

All functional requirements were validated successfully, and the system behaved as expected under all tested scenarios.

User Acceptance Testing (UAT)

User Acceptance Testing was conducted to verify that the solution satisfies business requirements and provides a satisfactory user experience.

The UAT process included:

- Request submission testing.
- Workflow validation.
- Approval verification.
- Catalog Task generation.
- Notification verification.
- Request lifecycle tracking.
- Update Set migration testing.

The testing confirmed that all business requirements were fulfilled successfully.

UAT Summary

- Total Test Cases Executed: **10**
- Passed: **10**
- Failed: **0**
- Success Rate: **100%**

Defect Analysis

During testing, a few minor issues were identified and resolved before deployment.

The major defects included:

- Catalog variables not mapping correctly to Requested Item (RITM).
- Catalog Task assignment issue.
- Missing mandatory field validation.
- Email notification delay.

All identified defects were corrected through updates to catalog variables, assignment rules, UI Policies, Business Rules, and notification configurations.

No critical or high-severity defects remained after testing.

Performance Testing

Although the project does not involve high-volume transaction processing, performance testing was carried out to verify workflow execution efficiency and system responsiveness.

The following performance parameters were evaluated:

Workflow Execution

The approval workflow executed successfully without delays.

Form Loading Time

The Service Catalog form loaded quickly and responded efficiently during user interaction.

Request Processing

Software requests were processed correctly from submission to completion with minimal processing time.

Notification Delivery

Email notifications were generated automatically after request submission, approval, task assignment, and request completion.

Database Operations

REQ, RITM, and SCTASK records were created and updated successfully without data inconsistency.

Reliability Testing

Reliability testing confirmed that the application consistently performed the intended operations without failure.

The system successfully maintained:

- Accurate request information.
- Correct workflow execution.
- Reliable approval routing.
- Consistent Catalog Task generation.
- Proper request closure.

Security Validation

Basic security validation was performed to ensure authorized access to application resources.

The following security mechanisms were verified:

- Role-Based Access Control (RBAC).
- Authenticated user access.
- Approval authorization.
- Mandatory field validation.
- Business Rule validation.
- Controlled access to Service Catalog.

Testing Summary

The Software Installation Request Automation solution successfully passed all functional and user acceptance tests. The workflow, approvals, notifications, catalog tasks, and request tracking operated correctly throughout the testing process. Minor defects identified during testing were resolved before deployment, resulting in a stable, reliable, and production-ready application.

The testing phase confirmed that the developed solution satisfies all functional and non-functional requirements while providing an efficient, secure, and user-friendly software request management process.





This screenshot shows the 'Discover' application in ServiceNow, displaying a table of discovered assets. The table includes columns for 'Name', 'IP Address', 'Vendor', 'Model', 'Serial Number', 'OS', and 'Status'.

Name	IP Address	Vendor	Model	Serial Number	OS	Status
10.10.10.10	10.10.10.10	HP	ProDesk	1000000000000	Windows	Active
10.10.10.11	10.10.10.11	HP	ProDesk	1000000000000	Windows	Active
10.10.10.12	10.10.10.12	HP	ProDesk	1000000000000	Windows	Active
10.10.10.13	10.10.10.13	HP	ProDesk	1000000000000	Windows	Active
10.10.10.14	10.10.10.14	HP	ProDesk	1000000000000	Windows	Active
10.10.10.15	10.10.10.15	HP	ProDesk	1000000000000	Windows	Active
10.10.10.16	10.10.10.16	HP	ProDesk	1000000000000	Windows	Active
10.10.10.17	10.10.10.17	HP	ProDesk	1000000000000	Windows	Active
10.10.10.18	10.10.10.18	HP	ProDesk	1000000000000	Windows	Active
10.10.10.19	10.10.10.19	HP	ProDesk	1000000000000	Windows	Active
10.10.10.20	10.10.10.20	HP	ProDesk	1000000000000	Windows	Active

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10.10.10.12	10.10.10.12	HP	ProDesk	1000000000000	Windows	Active
10.10.10.13	10.10.10.13	HP	ProDesk	1000000000000	Windows	Active
10.10.10.14	10.10.10.14	HP	ProDesk	1000000000000	Windows	Active
10.10.10.15	10.10.10.15	HP	ProDesk	1000000000000	Windows	Active
10.10.10.16	10.10.10.16	HP	ProDesk	1000000000000	Windows	Active
10.10.10.17	10.10.10.17	HP	ProDesk	1000000000000	Windows	Active
10.10.10.18	10.10.10.18	HP	ProDesk	1000000000000	Windows	Active
10.10.10.19	10.10.10.19	HP	ProDesk	1000000000000	Windows	Active
10.10.10.20	10.10.10.20	HP	ProDesk	1000000000000	Windows	Active

This screenshot shows the 'Discover' application in ServiceNow, displaying a form for asset discovery. The form includes fields for 'Name', 'IP Address', 'Vendor', 'Model', 'Serial Number', 'OS', and 'Status'.





CHAPTER 8

ADVANTAGES & DISADVANTAGES

The **Software Installation Request Automation using ServiceNow** project provides an efficient and standardized approach for handling software installation requests within an organization. By automating the complete request lifecycle, the solution minimizes manual intervention, improves operational efficiency, and enhances the overall user experience. However, like any software solution, it also has certain limitations.

Advantages

The major advantages of the proposed solution are as follows:

1. Process Automation

The entire software installation request lifecycle is automated, from request submission to approval, task assignment, fulfilment, and closure, reducing manual effort and improving operational efficiency.

2. Faster Approval Process

The workflow automatically routes requests to the appropriate approvers, significantly reducing approval delays compared to traditional email-based processes.

3. Improved User Experience

Employees can easily submit software requests through the ServiceNow Service Portal and track the status of their requests in real time.

4. Data Accuracy and Integrity

Mandatory field validation using UI Policies and Business Rules ensures that only complete and valid requests are submitted, improving data quality.

5. Automatic Task Assignment

Catalog Tasks are automatically generated and assigned to the appropriate Software Support Team, eliminating manual task allocation.

6. Real-Time Request Tracking

The use of REQ, RITM, and SCTASK tables enables users and administrators to monitor the complete request lifecycle.

7. Automated Notifications

Email notifications are automatically sent during request submission, approval, task assignment, and request completion, improving communication among stakeholders.

8. Secure Access Control

Role-Based Access Control (RBAC) ensures that only authorized users can submit, approve, or manage software requests.

9. Scalability

The solution can be easily extended to automate other IT services such as hardware requests, access requests, VPN requests, and employee onboarding.

10. Easy Maintenance

The use of ServiceNow Update Sets enables easy migration of configurations between development, testing, and production environments.

Disadvantages

Although the proposed solution provides significant benefits, it has a few limitations.

1. Dependency on ServiceNow Platform

The application can only be used within the ServiceNow environment and requires access to a ServiceNow instance.

2. Initial Configuration Effort

Setting up catalog items, workflows, UI Policies, Business Rules, notifications, and approvals requires careful planning and configuration.

3. Training Requirement

New users and administrators may require basic training to effectively use and manage the ServiceNow platform.

4. Customization Complexity

Advanced business requirements may require additional customization using ServiceNow scripting or integration capabilities.

5. Internet Dependency

Since the application is cloud-based, users require a stable internet connection to access the Service Portal and submit requests.

6. Licensing Cost

Organizations must maintain a valid ServiceNow license to utilize the platform and its advanced IT Service Management features.

CHAPTER 9

CONCLUSION

The Software Installation Request Automation using ServiceNow project successfully demonstrates how ServiceNow's Service Catalog and workflow automation capabilities can streamline the software request management process within an organization. The solution effectively replaces traditional manual request handling methods with an automated, standardized, and transparent workflow.

The developed system enables employees to submit software installation requests through the Service Portal by providing essential details such as Software Name, Version, Business Justification, License Type, and Urgency. The system automatically validates user inputs, routes requests for approval, generates Catalog Tasks for the Software Support Team, sends email notifications, and tracks the complete request lifecycle using the REQ, RITM, and SCTASK tables.

Throughout the project, Agile development practices were followed to ensure systematic planning, implementation, testing, and deployment. Functional testing and User Acceptance Testing (UAT) confirmed that all project requirements were successfully implemented, with every test case passing successfully. The automated workflow reduced manual effort, improved approval efficiency, enhanced request visibility, and increased overall user satisfaction.

The project also demonstrated the effective use of core ServiceNow features such as Service Catalog, Workflow Editor, Flow Designer, UI Policies, Business Rules, Notifications, Update Sets, and Role-Based Access Control (RBAC). These technologies collectively contributed to building a reliable, secure, and scalable IT service automation solution.

Overall, the project achieved its primary objective of automating software installation requests while improving operational efficiency and ensuring a consistent service delivery process. The developed solution provides a strong foundation for extending automation to other IT service requests, making it a valuable asset for organizations seeking to enhance their IT Service Management (ITSM) practices through ServiceNow.

CHAPTER 10

FUTURE SCOPE

The **Software Installation Request Automation using ServiceNow** project has successfully automated the software request process by providing a structured workflow for request submission, approval, fulfillment, and tracking. While the current solution meets the project's objectives, there are several opportunities for future enhancements to improve functionality, scalability, and user experience.

Future improvements include:

- **Automated Software Provisioning:** Integrate the solution with software deployment tools to automatically install approved software on users' devices after the approval process.
- **Integration with ServiceNow Integration Hub:** Use Integration Hub to connect with external systems and automate software deployment, license verification, and other IT operations.
- **Advanced Reporting and Dashboards:** Develop interactive dashboards to monitor request volume, approval trends, fulfillment time, and team performance for better decision-making.
- **Multi-Level Approval Workflow:** Implement additional approval levels based on software category, licensing cost, department, or organizational policies.
- **Artificial Intelligence Integration:** Incorporate AI-powered recommendations to suggest frequently requested software, detect duplicate requests, and provide intelligent assistance during request submission.
- **Mobile Access:** Enable users and approvers to submit requests, approve requests, and track request status through the ServiceNow Mobile application.
- **Expansion to Other IT Services:** Extend the existing workflow to support additional IT service requests such as hardware provisioning, user access requests, VPN requests, email account creation, and employee onboarding.

- **Enhanced Security:** Strengthen the application by implementing Multi-Factor Authentication (MFA), Single Sign-On (SSO), enhanced audit logging, and advanced role-based access controls.
- **Cloud and Third-Party Integrations:** Integrate the solution with enterprise platforms such as Microsoft Azure, AWS, Microsoft Teams, Slack, and other IT management tools to improve collaboration and automation.

Overall, the proposed future enhancements will transform the current solution into a comprehensive IT Service Management (ITSM) platform capable of supporting a broader range of enterprise IT services while improving operational efficiency, scalability, and customer satisfaction.

CHAPTER 11

APPENDIX

11.1 Source Code

Code for Business Rules:

```
if (current.state == 3) { // On Hold
    var inc = new GlideRecord('incident');
    inc.initialize();
    inc.short_description = 'License unavailable for ' + current.request_item;
    inc.description = 'Software request could not be fulfilled due to license shortage.';
    inc.caller_id = current.request.requested_for;
    inc.assignment_group.setDisplayValue('Software Support');
    inc.priority = 3;
    inc.insert();
}
```

11.2 GitHub Repository Link

<https://github.com/Abiramimaruthai>

11.3 Project Demo Link

<https://drive.google.com/file/d/1Pxt18WWTtIo7DRzhcG7J790uR54VaSGZ/view?usp=sharing>